POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2019
Programme: BE | Full Marks: 100
Course: Microprocessor | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the Instruction execution Cycle of microprocessor. 5
   b) Compare Von-Neumann's Architecture with Havard architecture. 5
   c) What are Flags? Write the function of all the flags present in 8085 microprocessor. 5
2. a) What are the different flags of 8085 microprocessor? Explain the addressing modes of 8085 microprocessor with examples of each. 2+6
   b) Write an ALP in 8085 to perform the following addition: 7
   $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2$
3. a) Define T-state. Draw the neat and labeled timing diagram for instruction MOV M, B. 8
   b) Write the function of DMA with suitable diagram and explain the execution of DMA in Master mode and Slave mode. 7
4. a) Differentiate between partial and complete address decoding. Design an address decoding circuits for two 16KB ROMs at address 2000H. 9
   b) Explain Polled and Vectored interrupt with suitable block diagram. 6
5. a) Draw the pin diagram of 8086 microprocessor and explain the different types of FLAGS of 8086. 7
   b) Write an 8086 ALP for MASM in DOS mode to print each word of a string in different lines. 8
6. a) What are the different pre-defined interrupts in 8086 microprocessor? Explain with the help of block diagram of 8086 IVT. 8
   b) How can you transfer the data between two computers without using modem? Draw and explain the interface with handshake Signal. 7
7. Write short notes on: (Any two)
   a) Explain the instruction : MVI, ANI, CMP,JNC,NOP
   b) RISC and CISC
   c) RS232 Standard

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